

ACADEMIC PAPER

Employment and economic growth dynamics in Odisha, India

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As the current phenomenon of “jobless growth” has gripped the Indian labor market, the subnational literature on how states have fared on employment generation in the post-liberalization period has been sparse. We examine the changing trends in employment and income among different sectors and subsectors of the economy of Odisha, an eastern Indian state, over the period 2011–2012 to 2017–2018 which is coincident to the Naveen Patnaik government which has been in power since March 2000 to date. Adopting the World Bank's Job Generation and Growth Decompositions tool (JoGGs) inspired by Shapley (1953. *Contributions to the theory of games*. Princeton University Press) decomposition method, we decompose growth across different sectors of its economy and examine how the per capita income growth is linked to changes in employment, output per worker, and population structure at the aggregate level and by sectors. We find there is a sharp decline in employment growth rate in the agriculture sector, a moderate growth in the mining sector, and a high growth only in the construction and a few service subsectors in Odisha. There has been significant output growth across industrial and service sectors and low growth in the agriculture-sector. Low employment creation in the agriculture sector and negative employment growth in manufacturing sector have had implications for seasonal migration and rural agrarian distress in the state. The dismal overall trend in the employment growth makes it imperative on the part of the government to undertake serious policy interventions in Odisha.

1 | INTRODUCTION

Indian economy has witnessed large variations in terms of employment and income growth over the last two decades. One of the stylized facts regarding employment and income growth in India during the post-liberalization period is that the economy has been able to surpass the “Hindu Growth” rate of around 3% seen during the 1980s to an average 8% growth rate during the first decade of the 2000s. However, the sectoral pattern of growth has been very uneven as there is a decline in the growth of the agricultural sector, moderate growth in the industrial sector, and the prominence of the service sector in the economy. In the national economy scenario, the economic debates in recent years have significantly focused on the employment–unemployment situation with most studies pointing toward the incidence of jobless growth in the Indian economy (Kapoor, 2017; Thomas, 2012). In this regard, when we discuss regional employment and income growth dynamics of the Indian states, the low growth of the workforce (i.e., low job creation)

turns out to be one of the contentious issues that needs comprehensive analysis. As the current phenomenon of “jobless growth” has gripped the Indian labor market, the subnational literature on how states have fared on employment generation in the post-liberalization period has been sparse. We want to address this shortfall.

Very few studies have examined the issue of employment growth dynamics at the subnational level in the Indian context. In this backdrop, there are three motivations as to why we undertook this study. First, years of good growth and a host of redistributive public welfare programs of the government have resulted in the claim that Odisha has successfully reduced poverty levels. However, how an indicator as crucial as employment has panned out in this journey of realizing improved social sector indices has not been subject to careful empirical examination in the literature. We fill this gap in this article. Second, we discern and analyze how the changes in employment levels have been attributed to the effects of different sectoral components. Third, we are also interested in the political economy implications, that is,

we want to know whether political regime persistence mattered in effecting the changes. The periodicity of the considerable economic progress in Odisha also begs the question about the incumbent Naveen Patnaik government's performance on employment so far. This article is a considerable addition to the empirical literature on employment dynamics at the subnational level in a developing country context. Studies like this are non-existent in the literature on Odisha, so the questions addressed in this study merited scientific enquiry.

In this context, we examine the changing trends in employment and income among different sectors and subsectors of Odisha economy over the period 2011–2012 to 2017–2018. Adopting the World Bank's Job Generation and Growth Decompositions tool (JoGGs) which had used Shapley (1953) decomposition method, we decompose growth across different sectors of its economy and understand how the per capita income growth is associated with changes in employment, output per worker, and population structure at the aggregate level and by sectors. We find that there is a sharp decline in employment growth in the agricultural sector accompanied by moderate growth in the manufacturing sector and high growth only in the construction and service sector. The period 2000–2018 coincident to the Naveen Patnaik government has seen good output growth across industrial and service sectors and negative growth for the agricultural sector. Low job creation in the farm sector and negative employment growth in manufacturing sector may have had implications for rising seasonal migration and rural agrarian distress in the state (also see Meher, 2019). In view of the dismal overall trend in employment growth, the government should implement corresponding policy interventions in the state.

The article is structured as follows. Section 1 presents the introduction section, Section 2 presents a review of literature, Section 3 includes the description of data sources and the empirical methodology. Thereafter, in Section 4, we present and discuss a comparative analysis of Odisha and Indian economy and other states in terms of employment situation. Section 5 presents the results of the JoGGs decomposition analysis. Section 6 concludes with a summary of the findings and policy implications.

2 | REVIEW OF LITERATURE

In recent years, Odisha has hogged limelight for its considerably impressive records on economic growth (e.g., see Sahu & Panda, 2018) and poverty alleviation and inequality reduction (e.g., see Panda & Padhi, 2020) vis-à-vis other major Indian states. In fact, an empirically rigorous study by Sahu and Panda (2018) found that the period contemporaneous to the Naveen Patnaik regime has seen higher growth performance of Odisha compared to its own history, the national economy, and 15 other major states. They found this incidence especially after 2003, a year of a structural break in growth, which happened to be the third year of the post-2000 Naveen Patnaik regime. Mr. Naveen Patnaik, the chairman of Biju Janata Dal, a regional political party, has been the Chief Minister of the state since March 2000 and has been only one of the three Chief Ministers in India's history to have won five consecutive terms in office (the other two being Jyoti Basu in West Bengal and Pawan Kumar Chamling in Sikkim). The state's performance

on poverty eradication has been spectacular: it has gone on from being the most underdeveloped state (see, e.g., Government of India, 2013) to be the state that has recorded the highest decline in poverty among the states (Panda & Padhi, 2020) and most impressive economic growth among the major states (Sahu & Panda, 2018).

Years of political stability and stupendous economic growth (Sahu & Panda, 2018) have meant that Odisha is no longer as poor as it was and is well on its path of economic progress. Some recent studies expound Odisha's impressive performance on several economic and human development indicators especially since the 2000s (Government of Odisha, 2015; Sahu & Panda, 2018; Samantaraya, Sahoo, Mallick, & Bhuyna, 2014). Examining the dynamic growth path of Odisha vis-à-vis its own recent history, the national economy, and other major states for the period 1980–2013, Sahu and Panda (2018) find an upward shift in Odisha's economic growth in the year 2003, after which the state has notched an average annual growth rate of about 8%. Their comparison of the state with the national economy as well as other 16 major states reveals a significantly positive differential trend break in Odisha's growth rate since 2003. The authors concluded that the Naveen Patnaik regime has coincided with the state's remarkable economic growth. It is worth mentioning that the period since 2000 to date has seen a regional political party (BJD) leading a persistent political regime in the state. The government led by Naveen Patnaik has pursued right-of-center public policies along with a host of populist public welfare schemes (Panda, 2019). Odisha moved from being the third poorest state in the pre-1990s reform period to the fourth poorest state in the post-reform period (Samantaraya et al., 2014). Odisha has become less poor in the last decade (2000–2010) despite its faring below the national average in several socioeconomic indicators (Samantaraya et al., 2014).

The decomposition analyses of Panda and Padhi (2020) revealed the redistributive changes in poverty and inequality over the period 1993–1994 and 2011–2012 across regions and social groups. They find that while Odisha has substantially reduced both rural and urban poverty incidence, it has not been able to register desirable subgroup specific changes. They also found that while inequality in urban regions rose, it declined in rural regions. While at the state-level, the within-group inequality has declined between 2004–2005 and 2017–2018 (coincident with the present Naveen Patnaik rule), the between-group inequality has increased during the period. With the two good tidings in place, it is natural to empirically enquire further as to how the Naveen government has fared in terms of job creation and employment growth in the state. The state government has claimed to have succeeded in its drive for skill development and job creation,¹ yet studies on the job creation and employment in the state have been sparse, if not nonexistent.

3 | EMPIRICAL ANALYSIS

3.1 | Data sources and methodology

We have used the data of two nationally representative large-scale employment surveys undertaken by the National Sample Survey

Organization (NSSO)² and published by Central Statistical Office (CSO). The two datasets pertain to the 2011–2012 and 2017–2018 rounds of the Employment and Unemployment Survey and the Periodic Labour Force Survey 2017–2018 (PLFS). Using the gross state domestic product (GSDP) data published by CSO (in base year 2011–2012 prices) and employment data from the NSSO, we analyze the long-term changes in income and employment in Odisha. The census adjusted weights have been used to execute relevant calculations from the NSSO employment and unemployment data sets.³ Following standard practice, we measure employment on the basis of the data on the usual principal and subsidiary status (UPSS) approach.⁴ We have done all the sectoral classifications as per the National Industrial Classification 2008 (NIC-2008) standards.⁵ The economy is broadly divided into three sectors such as Primary (Agricultural and allied activities and mining), Secondary (Manufacturing, Electricity and Construction), and Tertiary (Trade, hotel, restaurants; Transport storage and communication; Finance and real estate; Public Administration and Other services). Further, the secondary sector is divided into Manufacturing, Electricity and Construction sub-sectors. The service sector is further divided into sub-sectors such as trade, hotel, and restaurant (THR), transport, storage, and communication (TSC), financing, insurance, real estate, and business sector and community (FIRE), public administration (PUB), and other services (OTS).⁶ We consider population of the age group 15–65 years as the working age group in this study.

3.2 | Methodology: Analysis of Job Generation and Growth Decompositions tool

We have used a modified version of the Shapley (1953) decomposition method postulated by the Job Generation and Growth

Decomposition tool (JoGGs) developed by the World Bank (2012), in order to understand how the per capita income growth is linked to changes in employment, output per worker, and population structure at the aggregate level and by sectors in Odisha over the study period. The JoGGs methodology is a relatively simple additive method wherein the total change in per capita GSDP can be described in terms of the sum of the growth attributed to each of its components. It presents a unified way of looking at all the components (employment, capital, TFP, and sectoral reallocation of labor) and is very modest in terms of data requirements (Roy Choudhury & Chatterjee, 2015; Shorrocks, 2013; World Bank, 2012).

The Shapley decomposition method presented in a stepwise manner is utilized to decompose per capita GSDP growth into output per worker, employment, and capital. The JoGGs methodology has the following steps as shown in Chart 1.

Step 1 creates a profile of growth in per capita GSDP by decomposing it into changes in the employment rate, productivity, and demographics. In other words, it is about understanding the aggregate growth and productivity profile of growth. Step 2 decomposes changes in the employment rate into changes in employment by sectors. In other words, it is about understanding the role of each sector in employment generation. Step 3 decomposes changes in productivity into changes in output per worker within sectors and movement of workers between sectors. In other words, it is about decomposing the changes in output per worker by sectors and between and within components. Step 4 helps to further understand the role played by each sector in the overall change in output per worker. In other words, it shows an understanding the role of each sector in intersectoral shifts. The fifth and final step puts all the elements together, so as to see how each factor affected total per capita growth. In other words, it is the step of combining all the steps together.

Chart 1: Step wise Decomposition Approach (JoGGs)

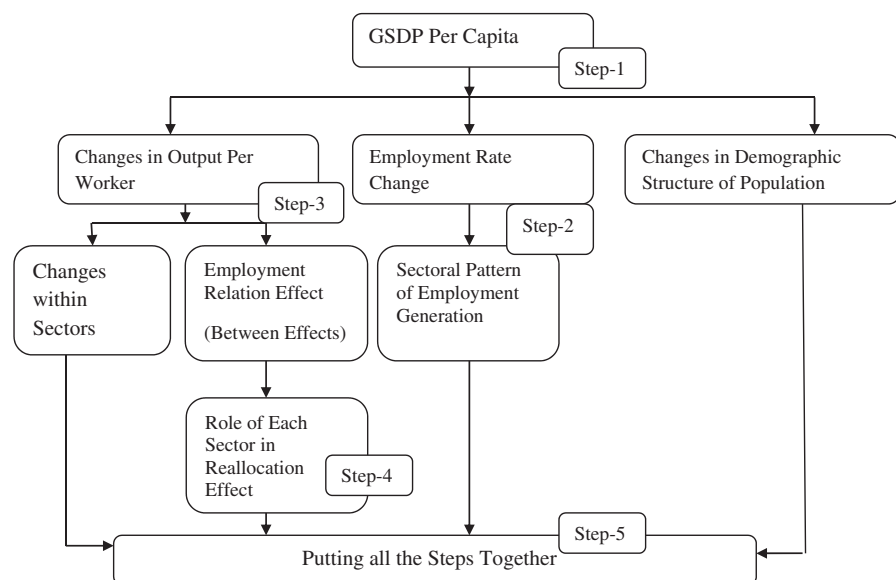


CHART 1 Step-wise decomposition approach of Job Generation and Growth Decomposition (JoGGs)

Source: The JoGGs Box Diagram to show the scheme of changes in growth of GDP per capita using the Shapley decomposition (World Bank 2012)

In our study, we have not included the total capital and their share in total income due to their unavailability in usable form in case of Odisha. So, the total factor productivity (TFP) has not been calculated in this analysis. Here, the total employment growth and income growth and their respective sectoral movements have been analyzed. For a concise explanation of the JoGGS methodology, please refer to Roy Choudhury and Chatterjee (2015, p. 587–592).

4 | ODISHA AND INDIAN ECONOMIES

In Table 1, we provide a brief comparison of the Indian states in terms of GSDP growth rate, employment growth rate, and employment elasticity. Table 1 shows that there exists a huge regional variation in terms of employment growth, GSDP growth and corresponding elasticity values. The employment growth rate is the highest in Madhya Pradesh (2.42%), followed by Jammu & Kashmir (1.95%) and

Chhattisgarh (1.72%). On the other hand, the highest negative employment growth rate has been seen in Nagaland (−5.36%), followed by Odisha (−1.86%) and Arunachal Pradesh (−1.55%). Corresponding to this, the employment elasticity (which is employment growth rate divided by the GSDP growth rate) is the highest for Meghalaya (0.525) followed by Jammu & Kashmir (0.359) and Chhattisgarh (0.309), whereas the highest negative employment elasticity has been found in Nagaland (−1.104) followed Arunachal Pradesh (−0.257) and Odisha (−0.241). These results give rise to a serious concern for Odisha, where even in the phase of high economic growth the state is witnessing a negative employment growth rate.

Similarly, Figure 1 reveals that Odisha economy has witnessed a stupendous increase in per capita GSDP (in rupees) during 2011–2012 to 2017–2018, although its growth performance was not comparable to the top performing states. Among the states, Sikkim, Haryana, and Gujarat have registered highest amounts of increase in per capita GSDP during the period.

	GSDP growth rate	Employment growth rate	Employment elasticity
Andhra Pradesh	8.00	−0.4	−0.049
Arunachal Pradesh	6.01	−1.55	−0.257
Assam	7.39	1.01	0.137
Bihar	6.31	−0.15	−0.023
Chhattisgarh	5.55	1.72	0.309
Gujarat	9.99	−0.53	−0.053
Haryana	8.53	0.27	0.032
Himachal Pradesh	7.11	−0.69	−0.097
Jammu & Kashmir	5.45	1.95	0.359
Jharkhand	6.29	−1.24	−0.197
Karnataka	9.38	−0.11	−0.012
Kerala	6.06	0.02	0.003
Madhya Pradesh	7.98	2.42	0.304
Maharashtra	7.2	−0.1	−0.013
Manipur	5.58	0.33	0.058
Meghalaya	2.97	1.56	0.525
Mizoram	12.59	−0.94	−0.075
Nagaland	4.86	−5.36	−1.104
Odisha	7.71	−1.86	−0.241
Punjab	5.87	−1.03	−0.175
Rajasthan	6.46	−0.71	−0.11
Sikkim	6.69	1.19	0.178
Tamil Nadu	6.82	0.08	0.012
Tripura	9.02	−0.3	−0.033
Uttar Pradesh	6.26	−0.54	−0.086
Uttaranchal	7.79	−0.7	−0.09
West Bengal	5.35	0.56	0.104
India	7.09	−0.004	−0.001

TABLE 1 GSDP growth, employment growth, and employment elasticity in Indian states

Abbreviation: GSDP, gross state domestic product.

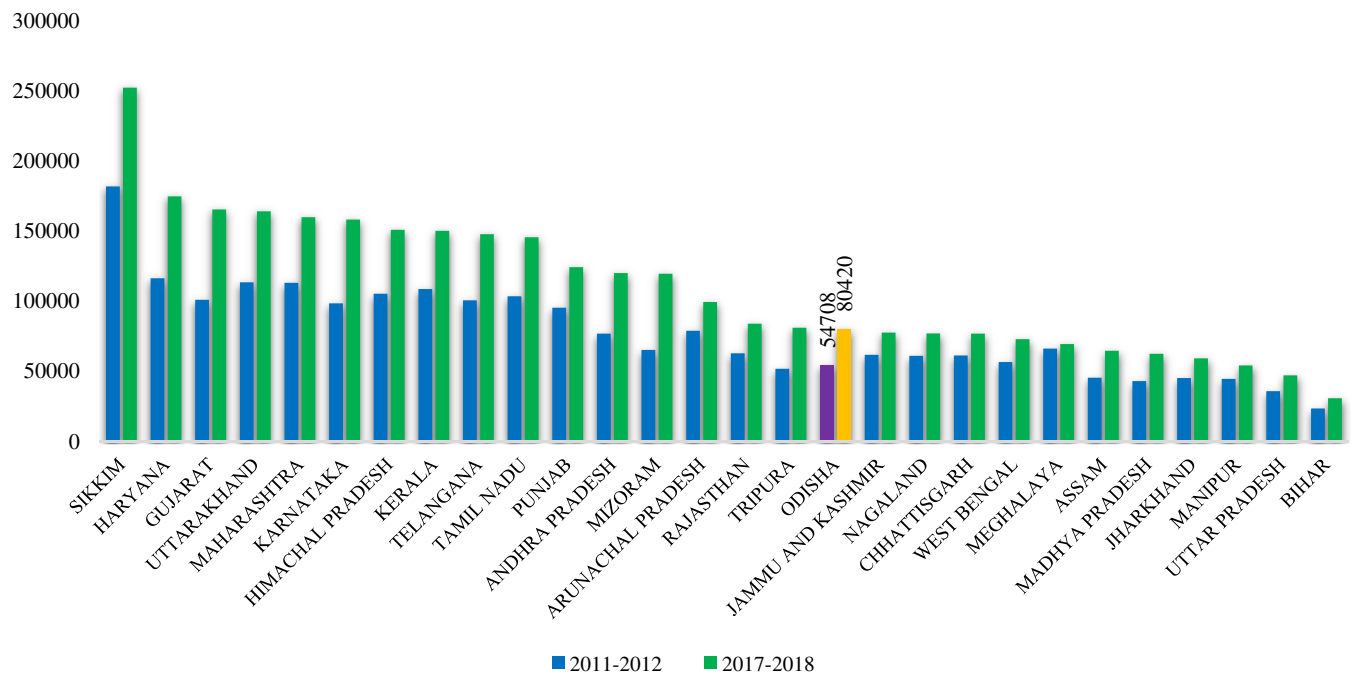


FIGURE 1 Per capita GSDP in rupees across states (2011–2012 prices). GSDP, gross state domestic product

TABLE 2 Sectoral distribution of GSDP and employment in Odisha

	GSDP share		Employment share	
	2011–2012	2017–2018	2011–2012	2017–2018
Agriculture	17.87	13.83	54.85	47.24
Mining	12.03	12.34	0.55	1.15
Primary	29.89	26.18	55.40	48.38
Manufacturing	18.69	21.39	9.74	7.71
Electricity	3.52	3.56	0.43	0.62
Construction	9.35	7.58	11.99	17.15
Secondary	31.57	32.53	22.16	25.49
THR	9.24	10.56	10.11	10.19
TSC	6.12	7.22	3.91	5.01
FIRE	11.37	11.68	0.66	0.74
PUB	3.92	4.78	1.32	1.30
OTS	7.89	7.05	6.44	8.89
Service	38.54	41.29	22.44	26.13
Total	100	100	100	100

Note: CSO, NSSO employment, and unemployment data of various rounds.

Abbreviations: Agri., agriculture; FIRE, finance and real estate; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication. These abbreviations will also hold in other places.

4.1 | Changing sectoral share of GDP and employment in Odisha

To understand the pattern of structural changes in Odisha, we estimate the change in sectoral shares of GSDP and employment during 2011–2012 to 2017–2018. Table 2 presents the sectoral composition of GSDP and employment for Odisha. As a standard practice, three

sectors—primary sector (Agriculture, forestry and logging and Fishing and Mining), secondary sector (manufacturing [both registered and unregistered], construction and electricity, gas and water supply), and service subsector have been analyzed.

There is an apparent shift in sectoral shares of GSDP during 2011–2012 to 2017–2018. During 2011–2012, the share of agricultural output accounted for 17.87% which has declined to 13.83% in

2017–2018. Secondary sector is almost stagnant in both time points (31.57% in 2011–2012 and 32.53% in 2017–2018). There is only increment in the manufacturing share in the secondary sector from 18.69 to 21.39% during this period, while the other two sectors show a declining trend. The service sector output has registered an increment from 38.54% of GSDP during 2011–2012 to 41.29% during 2017–2018. Like the national economy, Odisha has also witnessed a structural transformation in its economy in that both agriculture and services sectors have witnessed changes by about 3% points on opposite directions—agriculture sliding down and services sector moving up. This change has also reflected in the sectoral employment share.

By 2017–2018, the agriculture sector employment had declined by more than 7% points, whereas both the secondary and service sectors have increased by more than 3 and 4% points from their 2011–2012 levels. Among the subsectors, employment share of the construction sector has seen an impressive growth from about 12% in 2011–2012 to 17.15% in 2017–2018. Within the secondary sector, a higher employment share stems from the construction sector (17.15%) followed by the manufacturing sector (7.71%) in 2017–2018. Within the tertiary sector, the subsector “trade, hotel and restaurant” contributed the highest (10.19%) in 2017–2018. From the above analysis in Table 2, we infer that the sharp increase in the service sector output as a percentage of the state's GDP has not been accompanied by a corresponding drastic growth in its employment share during 2011–2012 to 2017–2018 in the state.

4.2 | Changing sectoral growth of GSDP, employment, and employment elasticity in Odisha

This pattern has adverse implications for reduction in poverty and inequality and generation of productive employment in Odisha. The declining trend in employment amidst a rising trend in economic growth is often depicted in employment elasticity which is the ratio of employment growth to that of growth of GSDP (Roy Choudhury and Chatterjee (2015). The employment elasticity can be calculated as

$$\text{Employment elasticity} = \frac{\% \text{Change in employment}}{\% \text{Change in GSDP}}$$

Table 3 depicts the employment and GSDP growth in the aggregate as well as at the sector and subsector levels during 2011–2012 to 2017–2018. Total employment has recorded a negative growth rate of −1.86% during 2011–2012 to 2017–2018, corresponding to this, the GSDP growth rate is 6.82%, respectively. Further, if we look at the employment growth across sectors, it is interesting to note a negative employment growth during 2011–2012 to 2017–2018 for the agricultural sector (−4.27%) in Odisha. It is quite significant as the state has been witness to regular out-migration to far off states in lean seasons that have low demand for farm labor.

This is a matter of concern given the secondary sector and service sector do not seem to be employing the surplus labor from agriculture. Among the subsectors, the mining sector (11%)

TABLE 3 GSDP growth rate, employment growth, and employment elasticity in Odisha during 2011–2012 to 2017–2018

	GSDP growth rate	Employment growth rate	Employment elasticity
Agriculture	2.36	−4.27	−1.810
Mining	7.28	11.00	1.511
Primary	4.48	−4.05	−0.904
Manufacturing	9.25	−5.61	−0.606
Electricity	7.01	4.45	0.636
Construction	3.15	4.17	1.326
Secondary	7.36	0.46	0.062
THR	9.21	−1.72	−0.187
TSC	9.81	2.26	0.231
FIRE	7.30	−0.12	−0.016
PUB	10.43	−2.05	−0.196
OTS	4.83	3.55	0.735
Service	8.05	0.66	0.082
Total	6.82	−1.86	−0.273

Note: CSO, NSSO employment, and Unemployment data of various rounds. Here, employment elasticity refers to employment growth rate divided by the GSDP growth rate.

Abbreviations: Agri., agriculture; FIRE, finance and real estate; GSDP, gross state domestic product; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

witnessed a higher employment growth rate, while manufacturing sector witnessed a highest negative employment growth (−5.61%) during this period. So, we can postulate that the overall growth of the secondary sector is due to the growth in the construction sector and electricity sector. Subsectors such as mining, electricity, construction, “transport, storage and communication,” and other services have shown a positive employment growth rate for Odisha during the period. In this context, it is interesting to note that the long-term deceleration of employment growth rate is accompanied by the acceleration of economic growth (see Table 3). Corresponding to this, public administration followed by the transport, storage, and communication and mining sectors showed a highest income growth.

The declining trend in employment volume is very well supported with the value of elasticity. Among the subsectors of industry and services sectors, almost all the sectors showed a positive increment. Among the subsectors of industry, growth in employment as well as its share in GSDP are found in case of the construction sector alone (1.326). The employment elasticity in the mining sector is 1.511. Among the constituent subsectors of services sector, the transport, storage, and communication followed by other services showed a high growth as compared to other subsectors of the services sector during 2011–2012 to 2017–2018. The value of the employment elasticity has been −0.273 for 2011–2012 to 2017–2018 in Odisha. Between sectors, the agriculture sector is showing a negative trend. In all the sectors, the employment elasticity is much lower in the recent periods (see Table 3).

5 | RESULTS AND INTERPRETATIONS OF JOGGs DECOMPOSITION

5.1 | Understanding the aggregate employment and productivity profile of growth in Odisha

The major summary statistics used for aggregate decomposition, that is, output, employment and population, employment share, output per worker, and share of working age population in Odisha are presented in Table 4 which displays that Odisha's GSDP per capita grew by 37.12% during 2011–2012 to 2017–2018. This GSDP per capita growth is accompanied by a rise in the growth of output per worker (66.25%) and a decline in total employment to –21.17% and an increase in the working age population by 3.12% during 2011–2012 to 2017–2018. The serious decline in employment rate is a matter of great policy concern. To comprehend the relative significance of each of the components, we implement the Shapley decomposition method as provided by the JoGGs tool.

Table 5 represents this decomposition in three components during 2011–2012 to 2017–2018. It is evident that growth in this period is predominantly influenced by the growth in output per worker. A significant part of growth in output per worker is owing to the change in the share of working age population to 14.68%. This result is very much in keeping with the demographic dividend benefitting the overall economic growth for the state.

The most striking aspect of this decomposition exercise relates to the contribution of the sharp deceleration in the employment rate (–77.81%) over the time period. The change in employment rate has become negative. This finding obviously points toward the lack of employment generation during this period when compared with the rising share of output per worker and the potential workforce owing to demographic dividend. The lower dependency increased per capita output since we are developing a counterfactual scenario wherein, we assume that all the working age adults faced the same employment rate and productivity in both periods (as mentioned in the World Bank JoGGs study). It is also important to note that a small change in the share of population of working age has huge effect on total growth (World Bank, 2012).

5.2 | Probing the differentials in employment generation across sectors

We also examine employment generation and growth across prominent sectors of the economy. Table 6 presents the summary statistics of employment across sectors. It shows that sectors such as agriculture, manufacturing, “trade, hotel and restaurant,” “finance and real estate,” and public administration, have witnessed an absolute reduction in total employment. All other sectors registered absolute growth in the number of people employed, with the mining subsector leading and being followed by electricity subsector in the secondary sector. The total employment has declined by 10.65% during 2011–2012 to 2017–2018. The working age population has declined by 21.17% during this period. Table 6 confirms that during this period, the growth in employment is positive in the secondary sector (mining, electricity, and construction) and service sector (transport, storage, and communication and other services) and the manufacturing (–37.59%) followed by the agriculture sector (–32.11%) showed a negative employment growth in this period. The overall declining contribution of employment rate is mainly attributed to the deceleration in manufacturing, agriculture, followed by public administration despite the modest acceleration in industry and service sector during the recent periods.

Corresponding to the results of Table 6, Table 7 presents the result of a similar decomposition analysis within sectors. We try to

TABLE 5 Decomposition of total growth in per capita GSDP in Odisha (%)

	Percentage Share
Growth linked to changes in output per worker	163.13
Growth linked to changes in employment rate	–77.81
Growth linked to changes in the share of population of working age	14.68
Total Growth in per capita GSDP (value added)	100

Source: Same as Table 2.

Abbreviation: GSDP, gross state domestic product.

TABLE 4 Macroeconomic indicators in Odisha, 2011–2012 to 2017–2018

	2011–2012	2017–2018	% Change
GSDP (value added) (in Rs. Cr)	220,231.69	327,152.27	48.55
Total population (in Lakh)	424.41	459.79	8.33
Total population of working age (in Lakh)	286.11	324.28	13.34
Total number of employed (in Lakh)	174.96	156.33	–10.65
GSDP (value added) per capita	51,890.81	71,153.17	37.12
Output per worker	125,877.93	209,273.94	66.25
Employment rate	61.15	48.21	–21.17
Share of population of working age	67.41	70.53	3.12

Note: Summarized from the table using JoGGs decomposition tool with the help of CSO and NSSO employment and unemployment data sets. The total population of working age belongs to the age group between 15 and 65.

Abbreviations: JoGGs, Job Generation and Growth Decompositions tool; GSDP, gross state domestic product.

TABLE 6 Employment by sectors of economic activity in Odisha in 2011–2012 to 2017–2018

	Total employment			Working age population		
	2011–2012(Lakhs)	2017–2018(Lakhs)	% Change	2011–2012	2017–2018	% Change
Agriculture	95.96	73.84	–23.05	33.54	22.77	–32.11
Mining	0.96	1.79	87.04	0.33	0.55	65.02
Manufacturing	17.04	12.06	–29.26	5.96	3.72	–37.59
Electricity	0.75	0.98	29.87	0.26	0.30	14.58
Construction	20.98	26.82	27.81	7.33	8.27	12.76
THR	17.68	15.93	–9.87	6.18	4.91	–20.48
TSC	6.84	7.83	14.37	2.39	2.41	0.91
FIRE	1.16	1.15	–0.70	0.41	0.35	–12.39
PUB	2.30	2.04	–11.67	0.81	0.63	–22.07
OTS	11.27	13.90	23.27	3.94	4.29	8.76
Total	174.96	156.33	–10.65	61.15	48.21	–21.17

Note: Same as Table 2.

Abbreviations: FIRE, finance and real estate; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

TABLE 7 Contribution of employment changes to overall change in employment rate and per capita GSDP (value added), Odisha 2011–2012 to 2017–2018

	Contribution of employment changes to overall change in employment rate		Contribution of employment changes to overall change in per capita GSDP	
	Contribution to change in total employment rate (percent points)	Percent contribution of the sector to total employment rate	Contribution to change in per capita GSDP (value added)	Percent of total change in per capita GSDP (value added)
Agriculture	–10.77	83.2	–12,470.15	–64.74
Mining	0.22	–1.68	252.11	1.31
Manufacturing	–2.24	17.3	–2,592.76	–13.46
Electricity	0.04	–0.3	44.34	0.23
Construction	0.94	–7.23	1,083.87	5.63
THR	–1.27	9.78	–1,465.64	–7.61
TSC	0.02	–0.17	25.21	0.13
FIRE	–0.05	0.39	–58.15	–0.3
PUB	–0.18	1.37	–205.83	–1.07
OTS	0.35	–2.67	399.54	2.07
Total	–12.94	100	–14,987.46	–77.81
	Total employment rate		Total contribution	

Abbreviations: FIRE, finance and real estate; GSDP, gross state domestic product; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

discover whether the so-called trope of jobless growth holds for all the sectors. With variations in the growth rate of each of the sub-sectors, such an exercise is worth attempting to unmask sector-specificity of the growth-employment nexus. This depicts a negative change in total employment rate during the period. The construction sector and most of the service sector components explain the increase in total employment growth rate during the period. This analysis reveals the negative contribution of employment in the agriculture sector to overall change in employment that has intensified in this period, with the contribution of the manufacturing sector

following suit. In fact, there is a notable increase in the share of the construction sector. The agriculture sector most significantly contributed to the decline in the employment rate. The second part of Table 7 represents the percentage contribution of the sectors to total growth in employment.

Apart from accounting for the share of employment, we further try to understand the response of changing growth in employment to growth in per capita output across sectors (see Table 7). Overall, the contribution of growth in employment to change in per capita GSDP is negative during this period. This overall negative contribution was

singularly owing to agriculture (83.20%) which has influenced all other sectors with a negative contribution toward growth in GSDP. In all other sectors, employment has contracted, thus having a negative effect on growth. This corresponds to overall negative effect on GSDP growth.

5.3 | Decomposing growth in output per worker within sectors and between sectors

Output per worker as presented in Table 8 depicts the differentials across sectors that have been varying over time. The notable feature of this analysis is that all other sectors have greater output per worker when compared with the agriculture sector and such differences seem to be widening over the period under consideration. The highest increment is in manufacturing sector where is a significant improvement around 140.32% during this period.

Table 9 depicts the contribution of each sector and intersectoral employment shift to the observed growth in total output per worker. Sectors such as manufacturing and “trade, hotel and restaurant” contributed positively to the increase in change in total output per worker during this time period (2011–2012 to 2017–2018). Of the total increase in the output per worker, the agriculture sector accounts for around 12.42% during this period. In sum, all the three sectors, namely, agriculture, manufacturing, and services, have witnessed increases in output per worker. The intersectoral labor shifts had a positive effect on output per worker (22.31%). This means that on average, labor moved from the low productivity sectors to sectors with relatively higher productivity. All this may be due to changes in the production structure that has taken place during the growth process.

TABLE 8 Changes in output per worker by sectors: Odisha 2011–2012 to 2017–2018

	2011–2012	2017–2018	% Change
Agriculture	41,001.30	61,289.32	49.48
Mining	2,764,833.83	2,253,478.47	–18.49
Manufacturing	241,514.50	580,406.17	140.32
Electricity	1,032,436.64	1,193,417.62	15.59
Construction	98,167.68	92,500.77	–5.77
THR	115,150.65	216,721.06	88.21
TSC	196,906.35	301,906.56	53.32
FIRE	2,159,338.41	3,318,830.53	53.70
PUB	374,568.46	769,088.02	105.33
OTS	154,180.53	165,959.75	7.64
Total output per worker	125,877.93	209,273.94	66.25

Note: Same as Table 2.

Abbreviations: FIRE, finance and real estate; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

5.4 | Understanding the sources of changes in total output per worker (net of intersectoral shifts) at the aggregate level

Table 10 depicts the contribution of changes in output per worker and intersectoral shift to total growth in per capita value added.

It is interesting to note in Table 10 that except the construction sector and mining sector, all other subsectors of the industry and services sectors contributed positively to total change in GSDP per capita with a positive intersectoral shift. This represents the shift of workers from low productivity sectors to sectors with relatively higher productivity. Except subsectors such as construction and mining, all other subsectors contributed positively to total percentage change in GSDP per capita, with a positive intersectoral shift. This analysis is in consonance with the evidence found in Roy Choudhury and Chatterjee (2015).

5.5 | Understanding the role of intersectoral employment shifts

An analysis of how changes in the share of employment in different sectors can explain the overall contribution of intersectoral shifts to per capita growth or output per worker. So, in this fourth step, we need to explain how changes in the share of employment in different sectors helps in explaining the overall contribution of intersectoral shifts to per capita growth (Roy Choudhury & Chatterjee, 2015). Many studies have

TABLE 9 Decomposition of output per worker into within sector changes in output per worker and intersectoral labor shifts for Odisha 2011–2012 to 2017–2018

	Contribution to change in total output per worker	Contribution to change in total output per worker (%)
Agriculture	10,355.44	12.42
Mining	–4,330.64	–5.19
Manufacturing	29,576.29	35.46
Electricity	848.06	1.02
Construction	–825.82	–0.99
THR	10,308.49	12.36
TSC	4,682.23	5.61
FIRE	8,110.96	9.73
PUB	5,166.01	6.19
OTS	903.09	1.08
Intersectoral shift	18,601.89	22.31
Total change in output per worker	83,396.00	100.00

Note: Same as Table 2.

Abbreviations: FIRE, finance and real estate; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

observed that structural transformation (labor movement from low productivity sectors to high productivity sectors) is an important feature of employment growth. If the share of employment in sectors that see above average productivity increases, it will shoot up overall productivity and the intersectoral labor shift term (Table 11).

On the contrary, labor movements out of sectors that see above average productivity leads to the opposite effect. Analogously, growth will be reduced if employment in sectors with below average productivity increases, while declines in the employment share should contribute positively to growth. Contribution of each sector to changes in output per worker is what is called the employment relocation effect. We further calculate the amount of total per capita growth that can be linked to relocation effects in each sector in Table 12. Mostly, the subsectors of the Services and Industry sectors showed a positive employment share and being high productivity sectors, contributed positively to intersectoral shifts. Hence, any shift of labor to these sectors positively influences per capita growth. Construction subsector had a negative employment shift share and being a high productivity sector in the recent periods, contributed negatively to intersectoral shifts.

5.6 | Putting everything together

Following the exploration of the varied aspects discussed above, the percentage contribution of each factor to total changes in

TABLE 10 Contribution of within sector changes in output per worker and intersectoral shifts to change in GSDP (value added) per capita in Odisha 2011–2012 to 2017–2018

	Contribution to change in GSDP (value added) per capita	Percent of total change in GSDP (value added) per capita
Agriculture	3,901.80	20.26
Mining	–1,631.73	–8.47
Manufacturing	11,143.99	57.85
Electricity	319.54	1.66
Construction	–311.16	–1.62
THR	3,884.11	20.16
TSC	1,764.21	9.16
FIRE	3,056.11	15.87
PUB	1,946.49	10.11
OTS	340.27	1.77
Intersectoral shift	7,008.97	36.39
Total contribution to change in per capita GSDP (value added)	31,422.60	163.13

Note: Same as Table 2.

Abbreviations: FIRE, finance and real estate; GSDP, gross state domestic product; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

GSDP per capita can be obtained. The last step in the decomposition exercise has been shown in Table 13. The sectoral contributions are decomposed into the following (Roy Choudhury & Chatterjee, 2015; World Bank, 2012):

- 1 Contribution of within sector changes in output per worker (first column),
- 2 Contribution of changes in employment (second column), and
- 3 Contributions of the sector to the intersectoral employment shifts.
- 4 The final column is the total effect of the sector. Overall, services, industry, and manufacturing had a positive contribution while the agricultural sector had a negative contribution.

The results essentially illustrate contribution in percentage terms and at constant 2011–2012 prices. The demographic component accounts for a positive 14.68% of the change in GSDP per capita growth rate during the period. The other 85.32% is explained by an increase in output per worker within sectors (126.74%), a decrease in the share of working age population employed (–77.81%), and a positive effect of labor relocation (36.39%) for the period 2011–2012 to 2017–2018. While looking across sectors, the most positive role was played by sectors such as manufacturing, mining, FIRE, TSC, THR. However, the agriculture sector did tot up a negative contribution during the period.

From Table 13, it is apparent that the negative contributions by the changes in employment in most of the sectors such as agriculture and manufacturing, and a negative intersectoral shift for manufacturing and construction, the massive positive growth linked with output

TABLE 11 The role of intersectoral employment shifts in Odisha 2011–2012 to 2017–2018

	Average output per worker	Change in employment share (percent points)	Sectoral contribution to intersectoral shift component
Agriculture	51,145.31	–0.076	8,863.94
Mining	2,509,156.15	0.006	14,017.79
Manufacturing	410,960.33	–0.020	–4,938.58
Electricity	1,112,927.13	0.002	1,840.97
Construction	95,334.23	0.052	–3,728.67
THR	165,935.86	0.001	–1.44
TSC	249,406.45	0.011	896.35
FIRE	2,739,084.47	0.001	1,896.27
PUB	571,828.24	0.000	–61.16
OTS	160,070.14	0.024	–183.58
Aggregate	167,575.94		18,601.89

Note: Same as Table 2.

Abbreviations: FIRE, finance and real estate; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

per worker in the sectors such as manufacturing, agriculture, THR, and FIRE, have been able to contribute a positive total growth in per capita income.

TABLE 12 Decomposition of intersectoral shifts in Odisha, 2011–2012 to 2017–2018

	Direction of employment share shift	Contribution to intersectoral Shifts (%)
Sectoral contributions		
Agriculture	–	47.65
Mining	+	75.36
Manufacturing	–	–26.55
Electricity	+	9.90
Construction	+	–20.04
THR	+	–0.01
TSC	+	4.82
FIRE	+	10.19
PUB	–	–0.33
OTS	+	–0.99
Total contribution of intersectoral shifts		100

Note: Same as Table 2.

Abbreviations: FIRE, finance and real estate; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

6 | CONCLUSION

As the current phenomenon of “jobless growth” has gripped the Indian labor market, the subnational literature on how states have fared on employment generation in the post-liberalization period has been sparse. We examine the changing trends in employment and income among different sectors and subsectors of Odisha economy over the recent period 2011–2012 to 2017–2018. Adopting the World Bank's Job Generation and Growth Decompositions tool (JoGGs) inspired by Shapley (1953) decomposition method, we decompose growth across different sectors of its economy and understand how the per capita income growth is linked to changes in employment, output per worker, and population structure at the aggregate level and by sectors in Odisha.

This analysis reveals the negative contribution of the employment in the agriculture sector to overall change in employment that has intensified over 2011–2012 to 2017–2018. In fact, there is a notable increase in the share of the construction sector. The performance of the agriculture sector is responsible for most of the decline during the time period. Positive increments are noted in all the sectors except only construction and FIRE sector during the period. On the whole, the contribution of growth in employment to change in per capita GSDP is negative during the period. This overall negative contribution was singularly owing to agriculture's bad showing and has been so intensive that it has turned all other sectors with a negative contribution toward growth in GSDP.

Sectors such as agriculture, manufacturing, and subsectors other than construction, electricity, mining, TSC, and OTS have witnessed downside in contribution to changes in employment growth, leading to a very precarious employment situation in the state. The

TABLE 13 Growth decomposition: Percent contribution to total growth in GSDP (value added) per capita, Odisha 2011–2012 to 2017–2018

	Contribution of within sector changes in output per worker (%)	Contribution of changes in employment (%)	Contributions of Intersectoral shifts (%)	Total (%)
Sectoral contributions				
Agriculture	20.26	–64.74	17.34	–27.14
Mining	–8.47	1.31	27.42	20.26
Manufacturing	57.85	–13.46	–9.66	34.73
Electricity	1.66	0.23	3.60	5.49
Construction	–1.62	5.63	–7.29	–3.28
THR	20.16	–7.61	0.00	12.55
TSC	9.16	0.13	1.75	11.04
FIRE	15.87	–0.30	3.71	19.27
PUB	10.11	–1.07	–0.12	8.92
OTS	1.77	2.07	–0.36	3.48
Subtotals	126.74	–77.81	36.39	85.32
Demographic component				14.68
Total				100
Total% change in value added per capita 2011–2012 to 2017–2018				37.121

Abbreviations: FIRE, finance and real estate; GSDP, gross state domestic product; OTS, other services; PUB, public administration; THR, trade, hotel, and restaurant; TSC, transport, storage, and communication.

intersectoral labor shifts had a positive effect on output per worker. This means that on average, labor moved from low productivity sectors to high productivity sectors. All this may be due to changes in the production structure that has taken place during the growth process.

We find there is a sharp decline in the employment growth in the agricultural sector, with negative growth in the manufacturing sector and high growth only in the construction and few service sectors. There has been a positive employment growth across industrial and service sectors and negative growth for the agricultural sector. Low job creation in the farm sector and negative employment growth in the manufacturing sector may have been influential factors in explaining the rise in seasonal migration and rural agrarian distress in the state. The dismal overall trend in employment growth makes it imperative on the part of the government to undertake serious policy interventions in Odisha.

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ENDNOTES

¹ Please see "Skill development: Odisha bags PM's Award for second time," May 3, 2018. <http://www.newindianexpress.com/states/odisha/2018/may/03/skill-development-odisha-bags-pms-award-for-second-time-1809604.html>, accessed on October 18, 2019.

² For detailed understanding of the concerned data sets, refer to the Report No. 554, 68th NSSO Employment and Unemployment Survey available at http://mospi.nic.in/sites/default/files/publication_reports/nss_report_554_31jan14.pdf, accessed on May 1, 2019.

³ For details regarding computation of population projections, kindly refer to the Report No. 554, 68th NSSO Employment and Unemployment Survey. The Census adjustment has been done on the basis of census and NSSO employment data sets. First, the weighted NSSO population figure has estimated from the concerned NSSO employment and unemployment rounds both for rural-urban and male and female differently after that the given figures are divided by the concerned census population figures. After getting the ratios, that are multiplied with the multiplier figures to get the census adjusted weights.

⁴ In this approach, the principal usual activity of the person is the status of activity on which a person spent relatively longer time of the preceding 365 days from the date of survey (refer to the Employment and Unemployment Survey report no. 554 stated in endnote 2). Accordingly, if the person was engaged for a relatively longer time during the past year in any one or more work related activities, a person is considered working or employed. For details, see the subsequent NSSO employment and unemployment reports, this analysis is directly taken from the NSSO employment and unemployment reports.

⁵ The sectoral classification has been done at one-digit level for broad seven groups which include agriculture, mining and manufacturing, construction and services, and others.

⁶ In dividing the service sector into different subsectors, we followed Roy Choudhury and Chatterjee (2015).

REFERENCES

- Government of India. (2013). *Report of the Committee for evolving a composite development index of states*. Chairman: Prof. Raghuram G. Rajan. Government of India, Ministry of Finance, New Delhi. September 2013. Retrieved from http://www.finmin.nic.in/reports/Report_CompDevState.pdf
- Government of Odisha. (2015). *Economic Survey—2014-15*. Retrieved from www.odisha.gov.in/pc/Download/Economic_Survey_2014-15.pdf
- Kapoor, R. (2017). Waiting for jobs. Working Paper No. 348, Indian Council for Research in International Economic Relations, New Delhi.

- Meher, S. K. (2019). Impact of distress seasonal migration from a socio-economic perspective: A case study of Nuapada district in Kalahandi Balangir Koraput region of Odisha. *The Indian Journal of Labour Economics*, 62, 763–782.
- Panda, S. (2019). Why does Odisha love Naveen Patnaik? *The New Indian Express*, May 24. Accessed from <http://www.newindianexpress.com/states/odisha/2019/may/24/why-does-odisha-love-naveen-1981119.html>
- Panda, S., & Padhi, B. (2020). The political economy of poverty and inequality in Odisha, India. Working paper.
- Roy Choudhury, P., & Chatterjee, B. (2015). Analyzing "jobless growth" in post-liberalisation India: a decomposition approach. *Indian Journal of Labour Economics*, 58, 577–608.
- Sahu, J. P., & Panda, S. (2018). Political regime persistence and economic growth in Odisha: An empirical assessment of the Naveen Patnaik rule. *Economics Bulletin*, 38(1), 610–622.
- Samantaraya, A., Sahoo, A. K., Mallick, A., & Bhuyna, B. (2014). Has Odisha become less poor in the last decade? *Economic & Political Weekly*, 49 (46), 62–67.
- Shapley, L. (1953). A value for n-person games. In H. W. Kuhn & A. W. Tucker (Eds.), *Contributions to the theory of games* (Vol. 2). New Jersey: Princeton University Press.
- Shorrocks, A. F. (2013). Decomposition procedures for distributional analysis: A unified framework based on the Shapley value. *Journal of Economic Inequality*, 11(1), 99–126.
- Thomas, J. J. (2012). India's labour market during the 2000s: Surveying the changes. *Economic and Political Weekly*, Vol. xlvii, No. 51.
- World Bank. (2012). JoGGs. Retrieved from http://siteresources.worldbank.org/INTEMPHAGRO/Resources/2743772-1239047170644/Guide_to_Growth_Employment_and_Productivity_Analysis.pdf

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